EART29102 Geospatial Techniques

Lecture 1: Course Overview and Introduction

Teaching staff:

Ray Burgess (co-ordinator)
Rufus Brunt
Peter Grinrod (Natural History Museum)
Ben Esse (practical classes)

Lecture outline

- What this course aims to achieve and how it is structured.
- The teaching and learning methods, feedback and assessment.
- What is a GIS and what is it used for?
- What are geospatial data?
- How data are represented (modelled) as vector and raster structures?
- A brief introduction to GIS software - ArcGIS Pro (practical).

Course Overview - Aims

1. Introduce and develop understanding of the principles and applications of geospatial techniques.
2. Show how different types of geospatial data can be collected (including remote sensing techniques).
3. Develop skills required to deal effectively with geospatial data in earth and environmental sciences through real world examples.
4. To be able to add, store, analyse and present geospatial data using a Geographical information systems (GIS).
5. Develop proficiency in using the GIS software ArcGIS Pro.

Intended Learning Outcomes (ILO's)

ILO 1: Understand what a GIS is and how it is structured.

ILO2: Define how spatial reference systems, datums, and projections are used in recording locations.

ILO3: Illustrate how geospatial data, at a wide range of scales, can be used to help solve scientific and societal problems.

ILO4: Choose appropriate geospatial techniques to solve a given scientific problem.

ILO5: Be competent in applying geospatial analysis and visualisation using ArcGIS Pro software.

Feedback and assessment

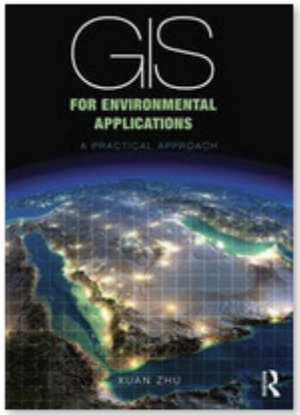
Formative

- Weekly short answer and multiple-choice quizzes based on lecture content (ILOs 1-3).
- Weekly feedback on ArcGIS Pro examples and workshops to support progress of case studies (ILOs 4-5)

- **Summative**
- Canvas test of 50 questions to demonstrate knowledge and understanding of GIS relevant to earth and environmental applications (summer exam period) – ILO's 1,2,3 (70%).
- ArcGIS Pro software project on selected case study to demonstrate competence of ArcGIS Pro and analysis of spatial data (30%). Submission at end of week 11 (30th April, 2026) (ILO's 4,5).

Teaching methods

- 1 hr lecture and 2 hr practical class each week.
- The lectures will provide the background theory and context for the practical exercises.
- The module will use ESRI ArcGIS Pro software in practical classes.
- Supported by on-line videos (mainly ESRI training videos) and core textbook.



Book

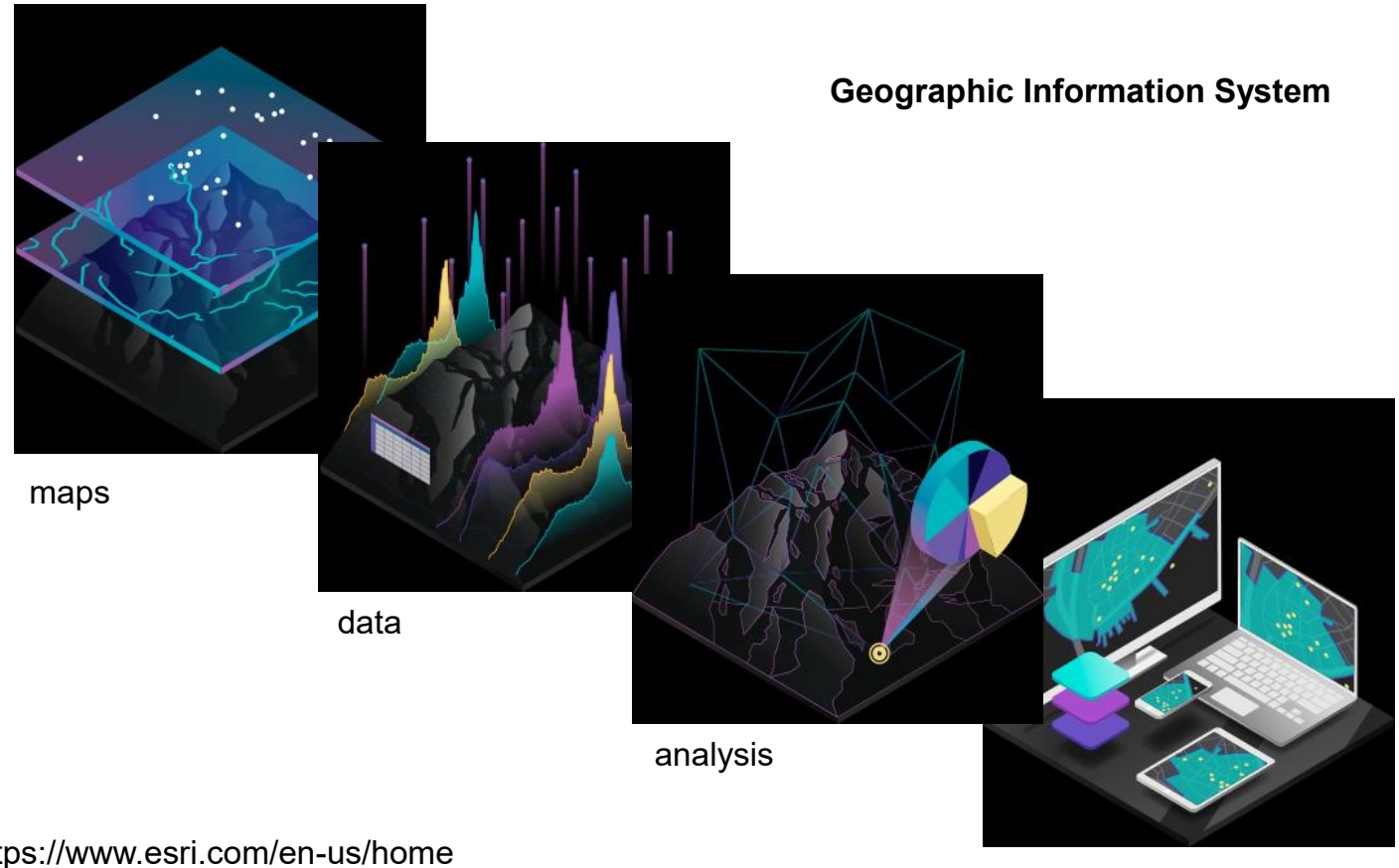
GIS for Environmental Applications

A practical approach

By Xuan Zhu

Edition	1st Edition
First Published	2016

Introduction to GIS



https://www.esri.com/en-us/home

Geographic Information System (GIS)

Geographic Information System (GIS) is a computer-based system for capture, management, storage, organization and analysis of spatially-referenced data, and the output of spatially-referenced information.

Data are a collection of related facts; the basic elements of information.

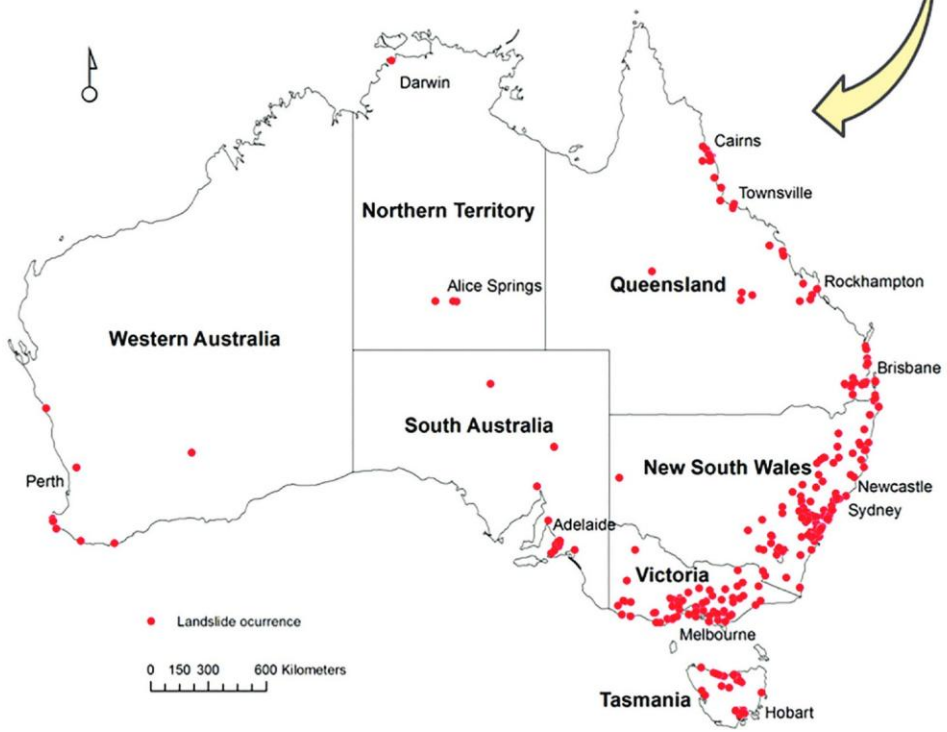
Information is data that have been processed to be useful. It provides answers to questions, used in decision-making.

Non-spatial data: a summary of the landslides occurred in Australia since 1900

TYPES OF LANDSLIDES	NO. OF OCCURRENCE
debris avalanche	2
debris fall	3
debris flow	26
debris slide	23
debris slump	14
debris topple	1
dry sand flow	1
earth block slide	4
earth fall	3
earth flow	27
earth slump	22
rock fall	111
rock slide	14
rock slump	8
rock topple	4
slump-earth flow	8
slump and topple	1
soil creep	3
wet sand or silt flow	1

Spatial data:
locations of the landslides occurred in Australia since 1900

DATE	LOCATION	TYPE	LONGITUDE	LATITUDE
23/03/1997	Trinity Beach, Cairns	debris fall	145.70	-16.79
4/02/2000	O'Connor, Kalgoorlie-Boulder area, WA	debris fall	121.48	-30.76
1/04/1911	Nisbet Range, 5 km E of Cairns	debris flow	145.80	-16.90
1/01/1925	Mt Ainslie, Canberra, ACT	debris flow	149.17	-35.27
1/08/1970	Plenty River, 10km west of New Norfolk	debris flow	146.94	-42.80
30/04/1988	Coledale	debris flow	150.93	-34.28
22/09/1989	Sulphur Creek, NW Tasmania	debris flow	146.02	-41.10
1/05/1996	Mt Mitchell, near Cunningham's Gap, Qld	debris flow	152.40	-28.07
19/01/1997	Killaloe, near Mossman, Qld	debris flow	145.38	-16.47
30/07/1997	Thredbo, Snowy Mountains	debris flow	148.31	-36.51
10/01/1998	Mt. Stuart, Townsville, Queensland	debris flow	146.78	-19.35
11/01/1998	Near Kelly St, Nelly Bay, Qld	debris flow	146.84	-19.16
7/08/1998	Below Seaforth Crescent,Sydney	debris flow	151.25	-33.80
17/08/1998	Wollongong	debris flow	150.90	-34.43
11/02/1999	Kuranda Range Road, Kuranda, Qld	debris flow	145.64	-16.82
11/02/1999	Barron Gorge Road, Cairns, Qld	debris flow	145.67	-16.87
3/03/1999	Great Western Tiers, Meander, Tasmania	debris flow	146.62	-41.75
23/10/1999	Nareena Hills, Wollongong	debris flow	150.83	-34.43
...	...	...	...	...



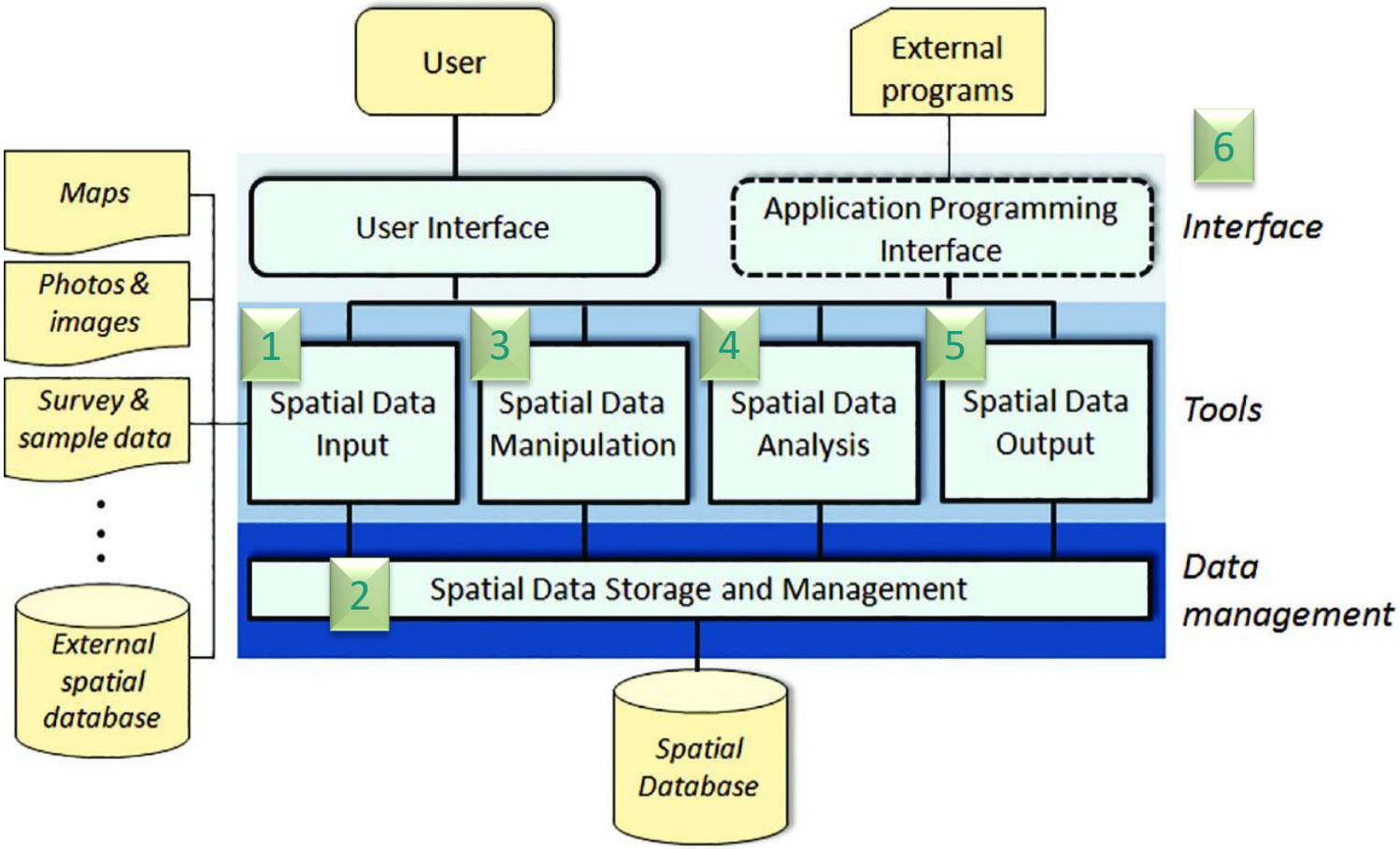
What does a GIS do?

- Capture data: add data from many sources to a GIS and create data from scratch.
- Store data: store and manage information about the real world in ways that make sense for an application.
- Query data: ask complex questions about features based on their attributes or their location and get quick results.
- Analyse data: can integrate multiple datasets to find features that meet specific criteria and create information useful for problem solving.
- Display data: display features based on their attributes, a powerful feature of any GIS.
- Present data: create and distribute high-quality maps, graphs, and reports to present analysis results in a compelling way to an audience.

Features of spatial data

- Spatial data are related using a coordinate system which defines spherical or planimetric coordinates, elevation, orientation etc.
- Observations and measurements are made on geographically distributed features such as plants, wildlife animals, rivers, etc. Each of these spatial objects is an entity that can be described by its attribute or property.
- Attributes or properties may include observable and/or measurable characteristics, e.g. Tree is observation, latitude and longitude define its location, while elevation, aspect, slope, solar radiation, soil type, etc are attributes specific to that location.

GIS functional components



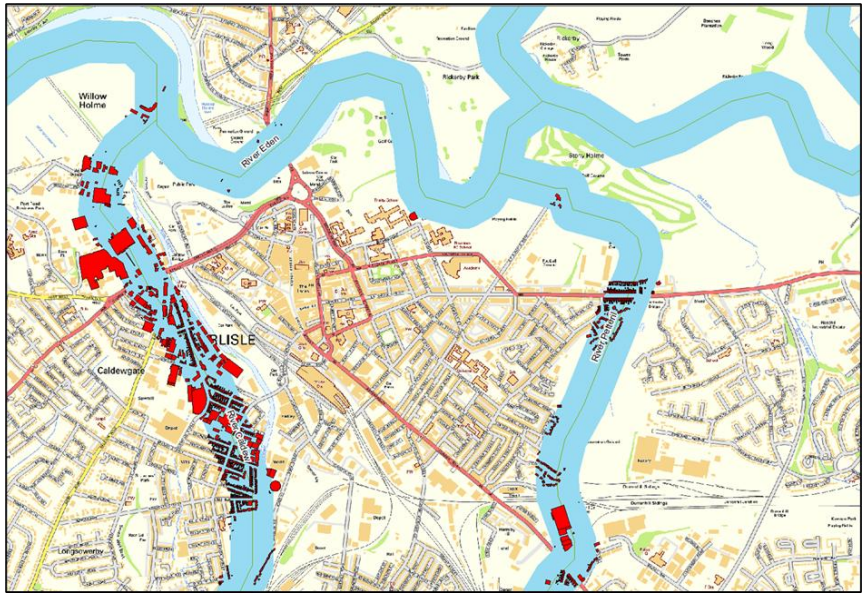
Spatial data analysis component

- A collection of tools for exploring patterns, associations and interactions of environmental phenomena as well as evidence of their changes through time and over space.
- Include spatial query, geometric measurement, buffering and proximity measurement, overlay analysis, surface analysis, network analysis, image analysis and spatial statistics.

Buffering and proximity measurement

- Functions used to calculate distances around (buffer) and from (proximity) a feature or features.

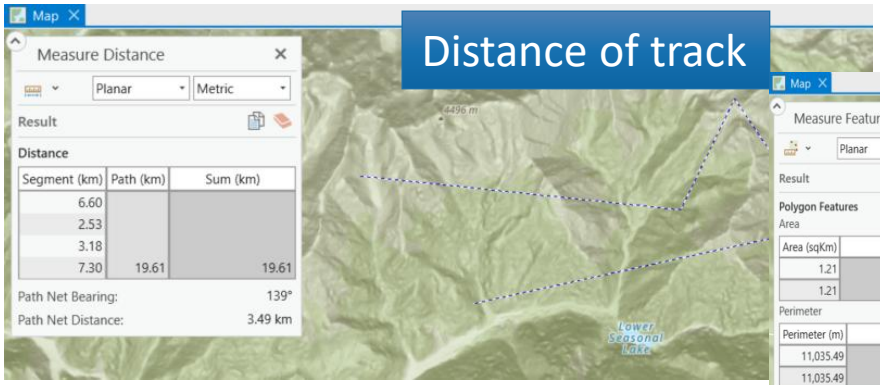
  e.g. use a 100 m buffer around rivers or calculate distances from a river to assess flood risk.

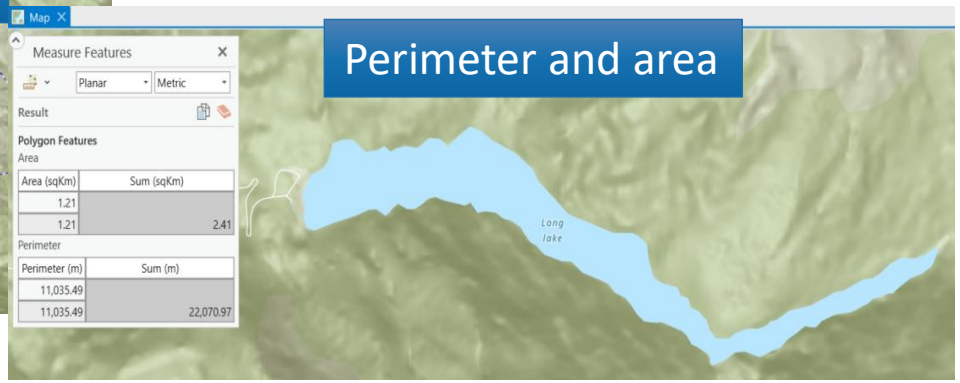


Buildings in red are in 100m buffer zone of rivers.

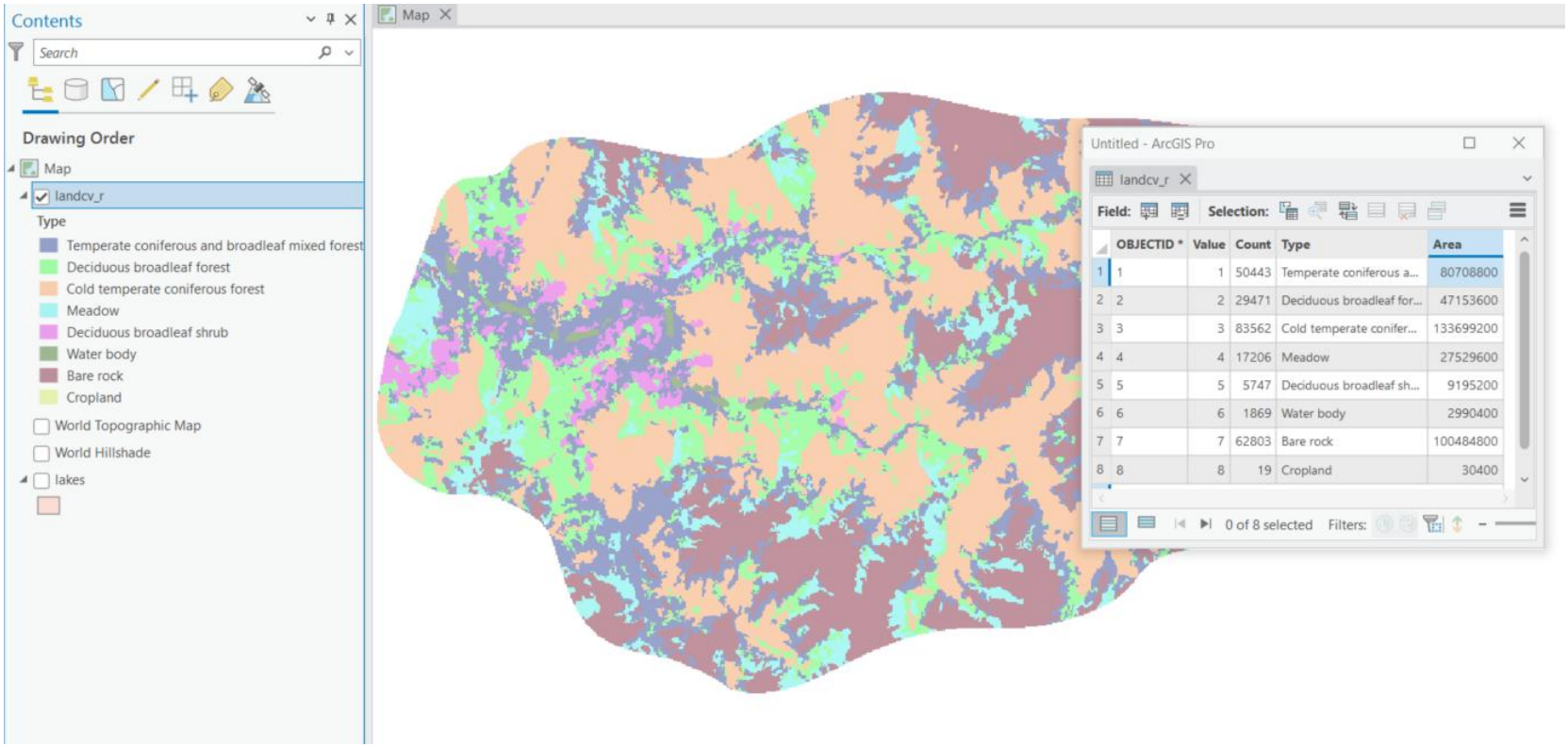
Geometric functions

- Calculate properties such as length, perimeter, area, shape, orientation, centroid and distances.
- Useful for landscape metrics, analysing landscape fragmentation and characterising landscape structures.



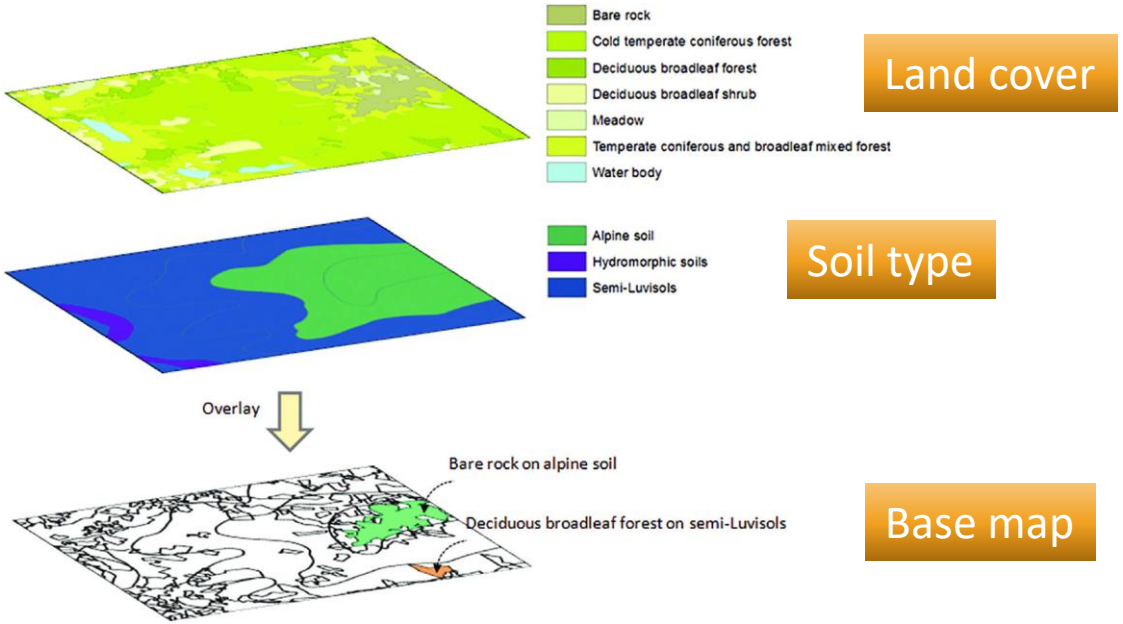


Layers and Attribute tables



Overlay analysis functions

- Stacking of layers of different themes based on common geography.
- Useful to identify spatial associations (e.g. overlaps) for optimal site selection or suitability modelling.



GIS uses layers of data

- GIS often combines information from different *layers.*
- Layers form a stack with multiple datasets found in x,y,z location.

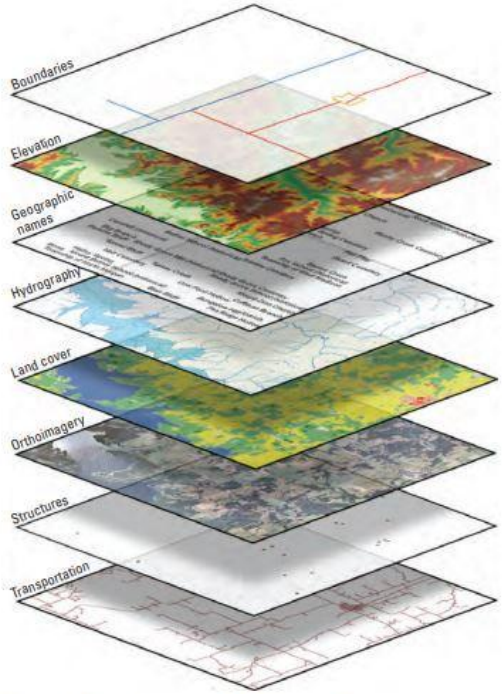


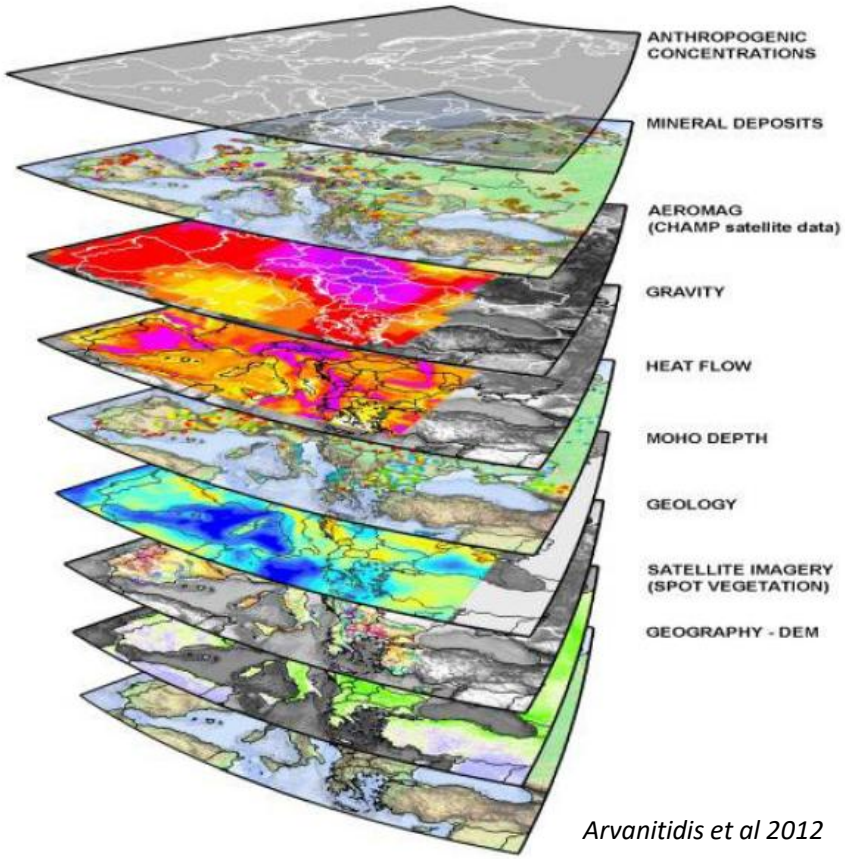
Figure 1. Eight base layers of *The National Map.*

USGS National Map provides access to base geospatial information to describe the landscape of the United States and its territories.

- Comprises 8 layers at 1:24,000 scale
- On-line National Map viewer https://apps.nationalmap.gov/viewer/

https://www.usgs.gov/programs/national-geospatial-program/national-map

Mineral deposit exploration



Arvanitidis et al 2012

- Background map
- Aerial photos/satellite images
- Environmental data
- Geological field data
- Geophysical data
- Geochemical data

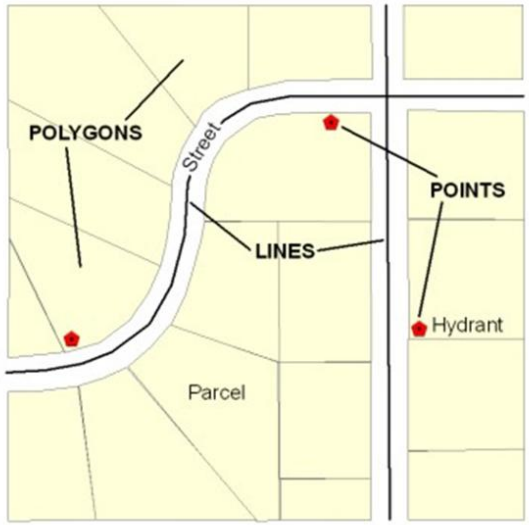
Spatial data models

The most common representation of spatial data that measures the landscape is using discrete data (vector model) and continuous data (raster models).



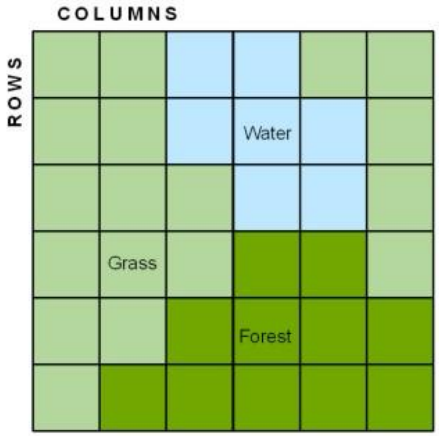
Vector data model

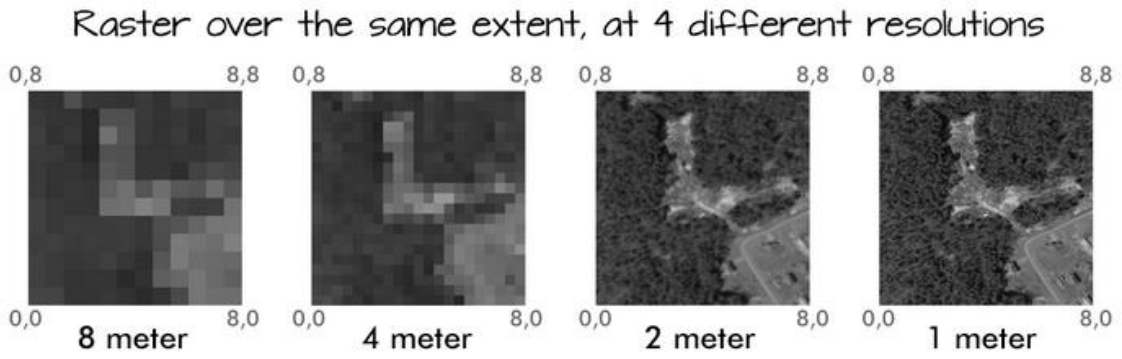
- Features that have definite boundaries and identities and a specific shape using points, lines, and polygons.
- Useful for storing data that have discrete boundaries, such as groundwater wells , streams, and lakes.



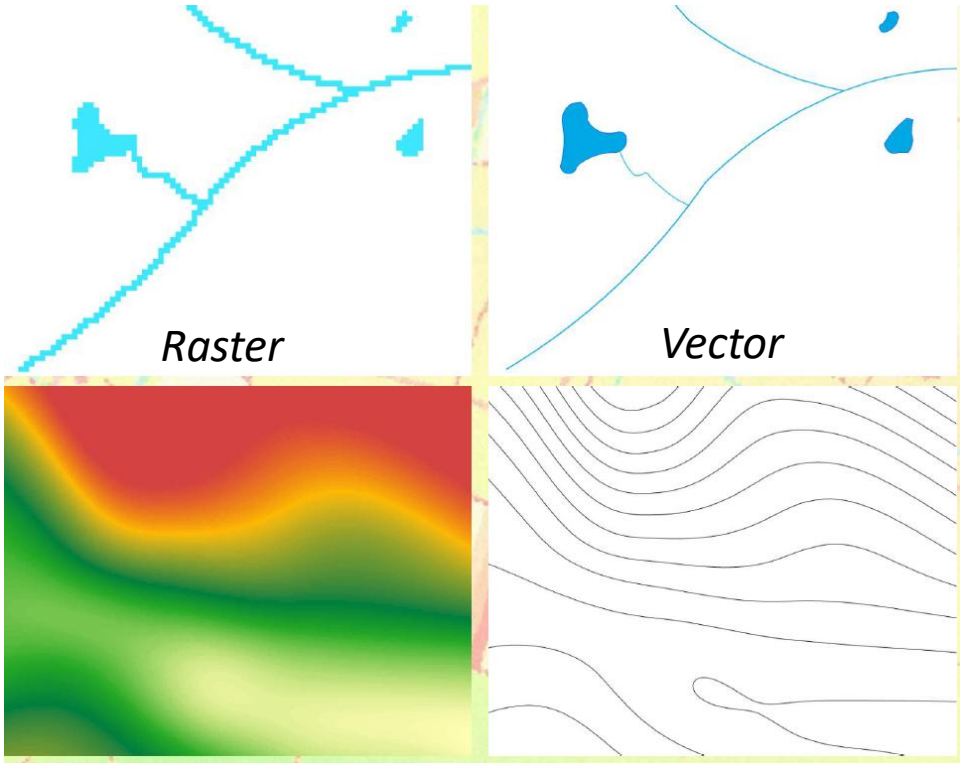
Raster model data

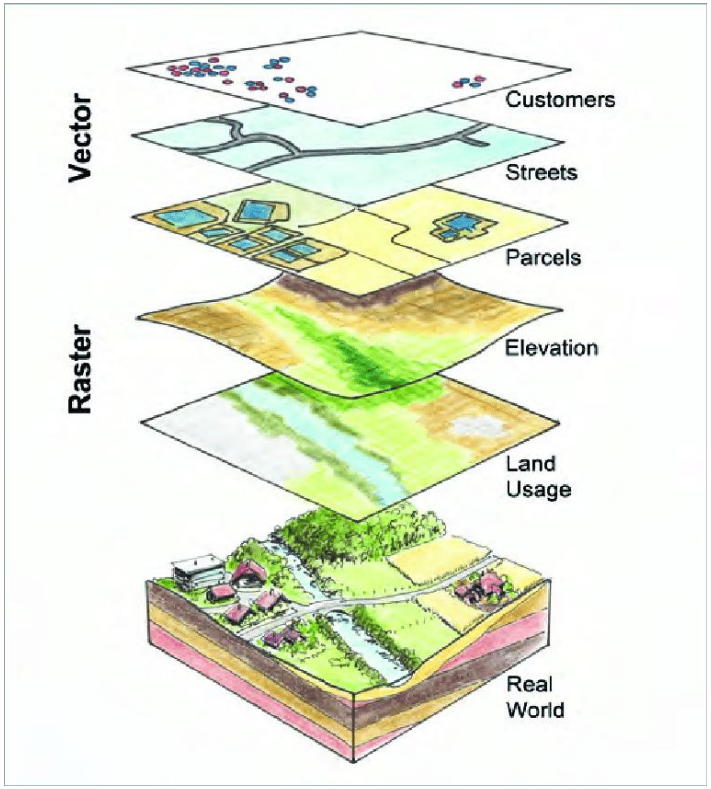
- Represent geographic features by dividing them into a discrete grid of cells (pixels).
- Useful for storing data that vary continuously, e.g. aerial photographs, satellite images, a surface of humidity, or digital elevation model (DEM).



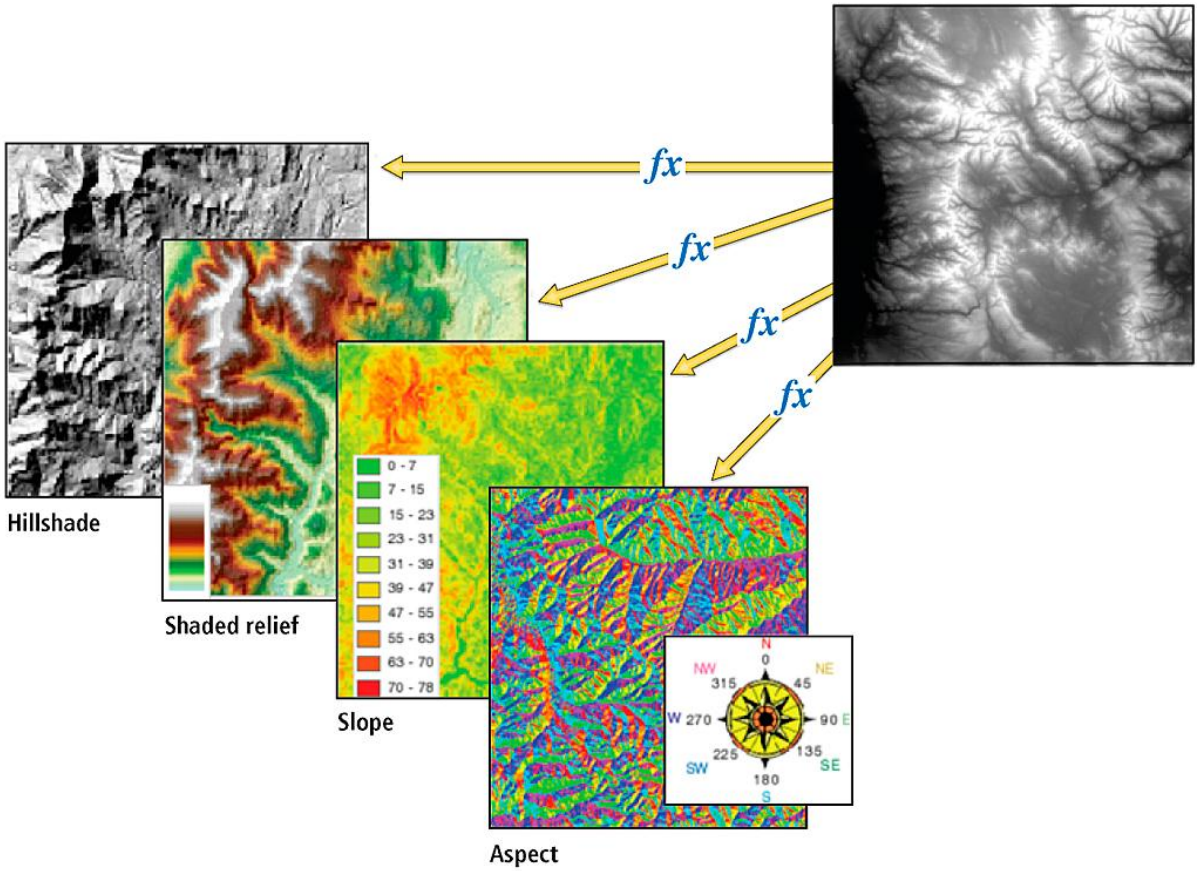


Raster versus Vector



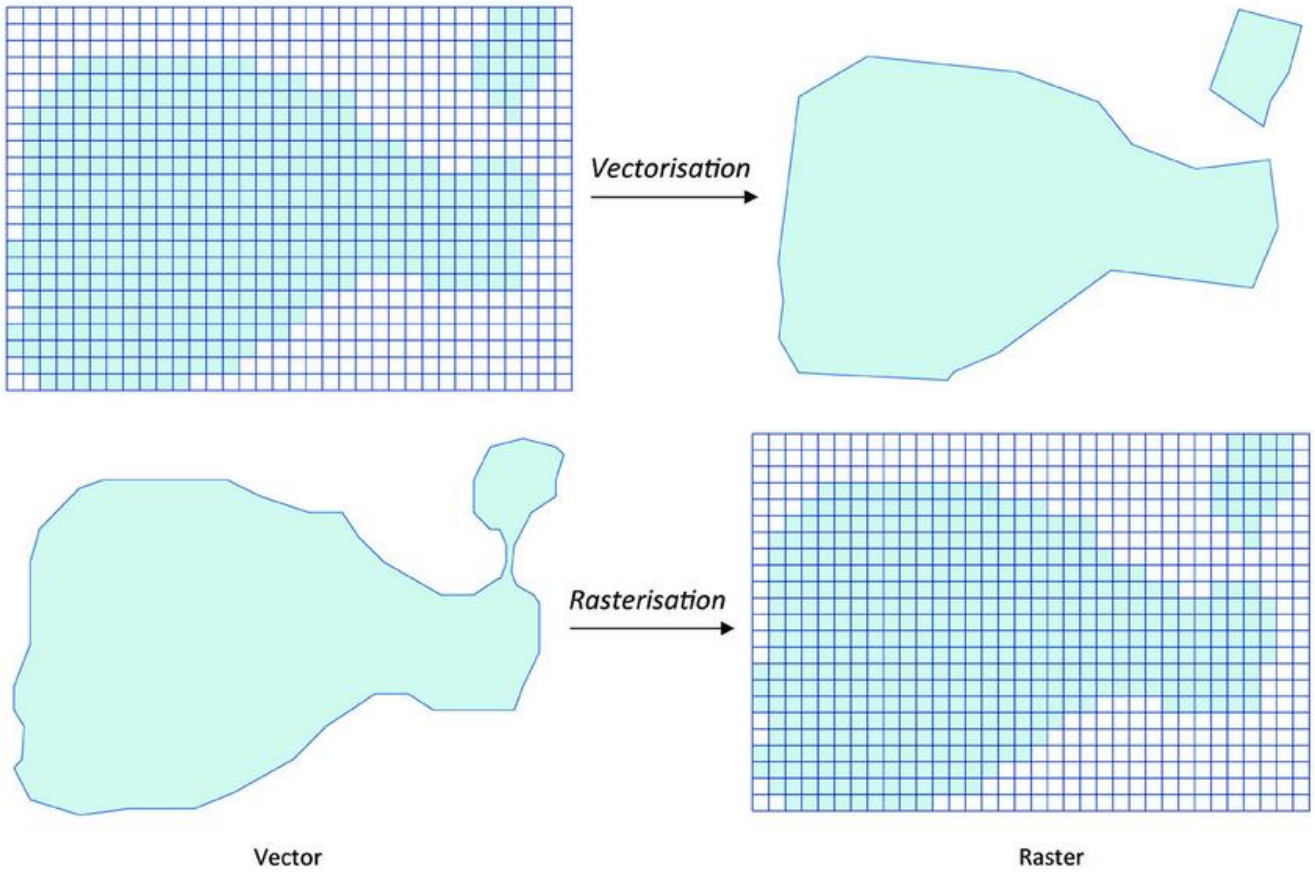


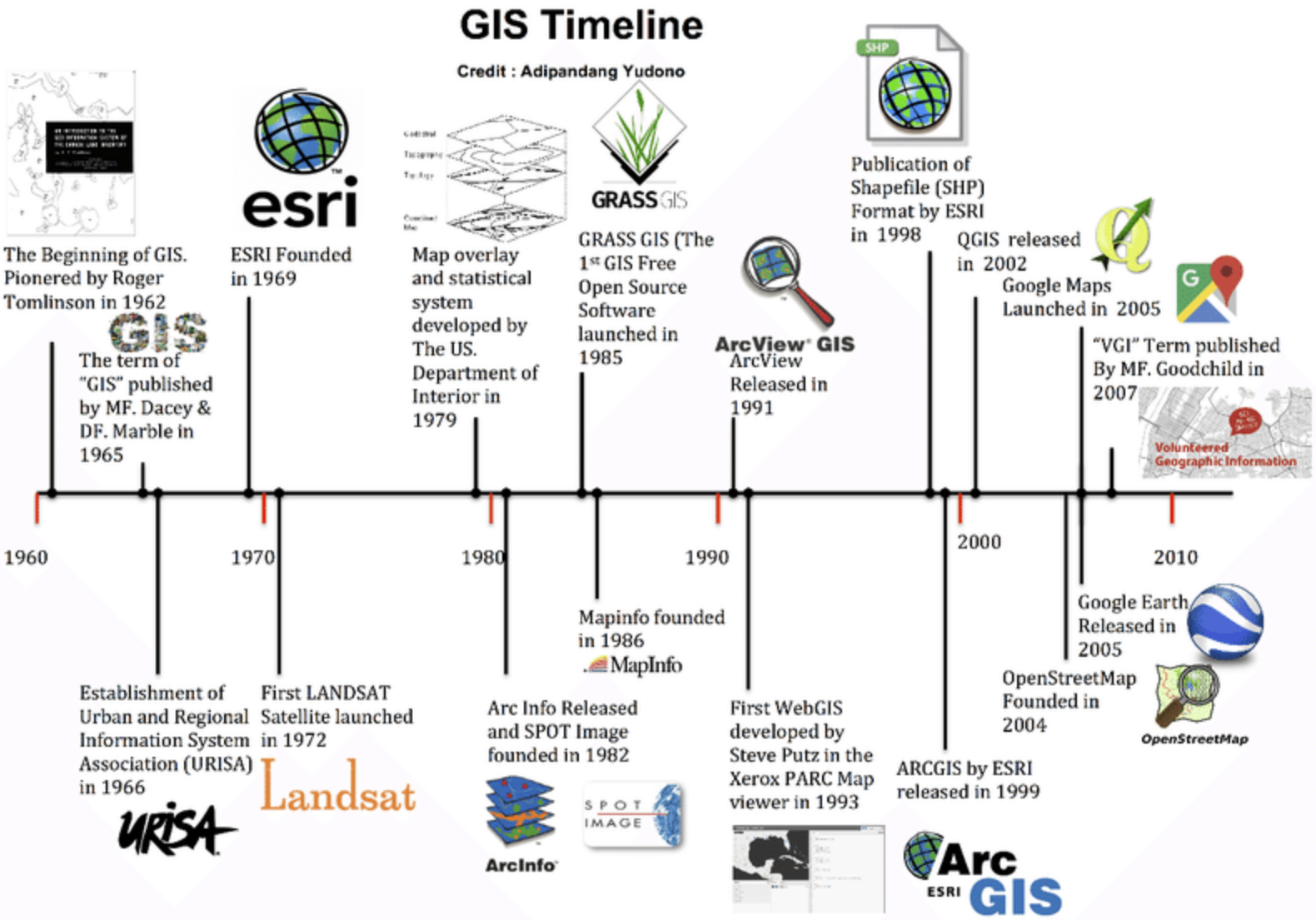
Digital Elevation Models (DEM)



https://www.esri.com/news/arcuser/0610/mosaicdataset.html

Spatial transformation



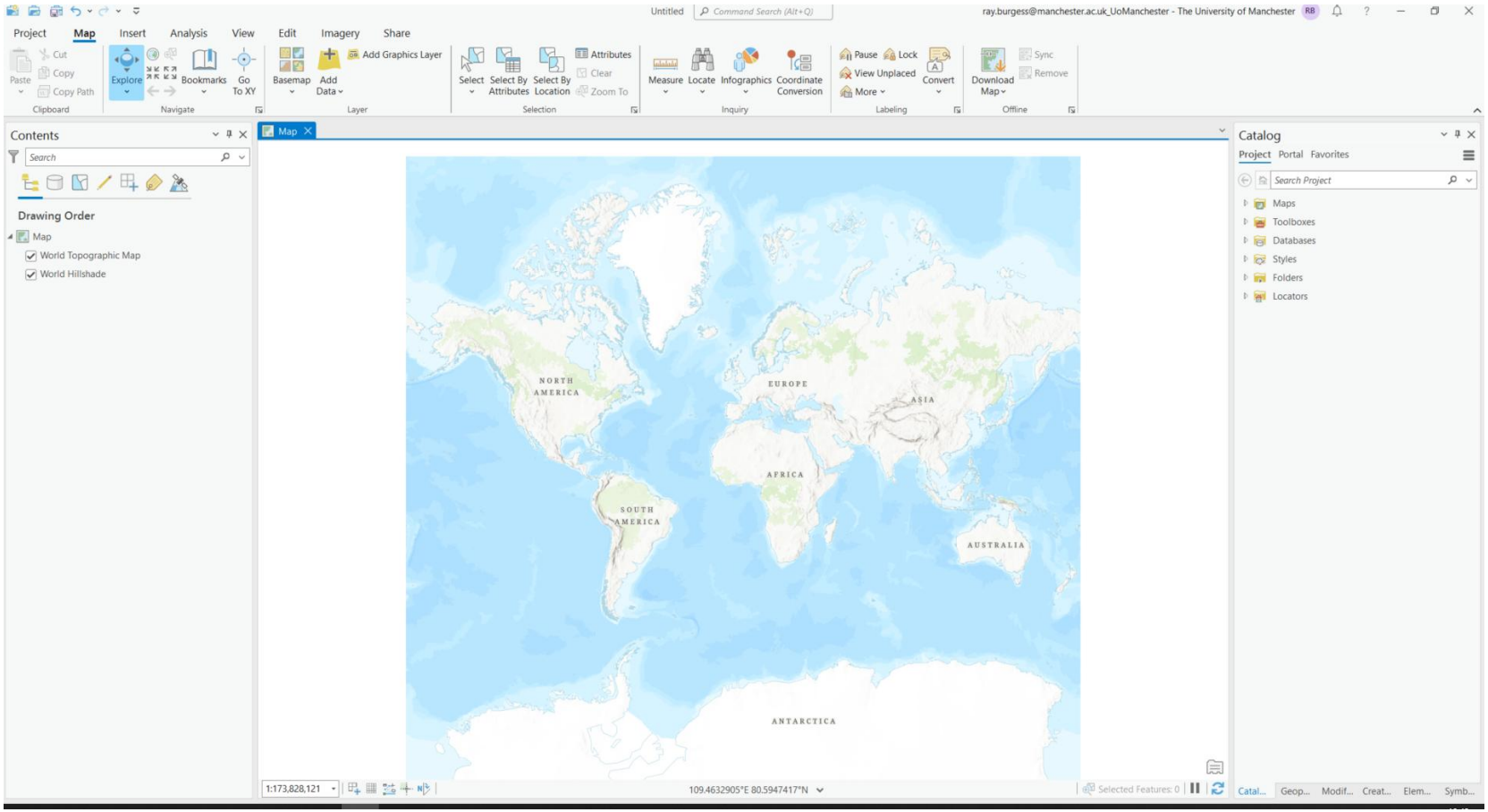


Yudono (2017) https://iopscience.iop.org/article/10.1088/1755-1315/70/1/012002/meta

Introduction to ArcGIS Pro

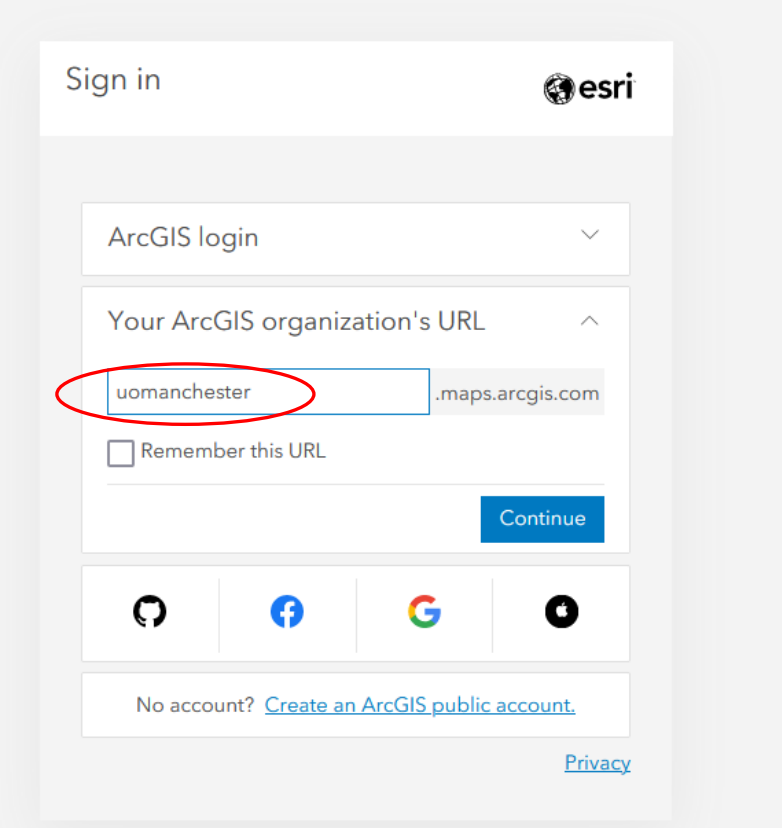
- ArcGIS Pro is an advanced GIS application from Esri.
- Consists of maps, scenes , layouts , data , tables, and different tools that are structured in a project.
- ArcGIS Online is a cloud-based system built to encourage the sharing of GIS data and other resources between organizations and users.
- Both ArcGIS Pro and ArcGIS online required an account from ESRI that provides access to more data and tools.
- ArcGIS Pro can also run offline. ArcGIS Pro has the capability of ArcGIS desktop (ArcMap, ArcCatalog, and ArcToolbox). It also enables Python and ModelBuilder.

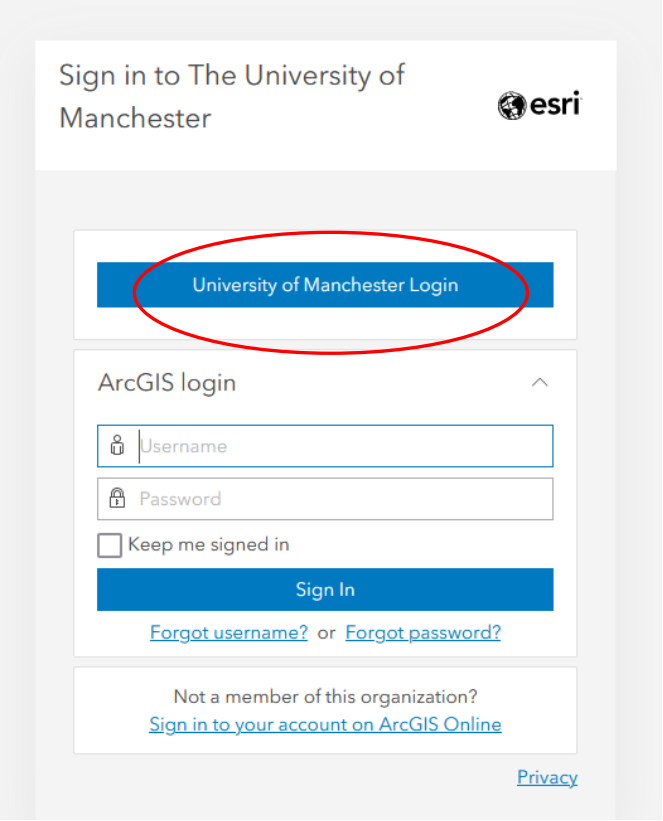
ArcGIS Pro


Project
Map
Insert
Analysis
View
Edit
Imagery
Share
Contents
Drawing Order
Map
World Topographic Map
World Hillshade
NORTH AMERICA
EUROPE
ASIA
AFRICA
SOUTH AMERICA
AUSTRALIA
ANTARCTICA
Catalog
Project
Portal
Favorites
Maps
Toolboxes
Databases
Styles
Folders
Locators

ArcGIS online - access

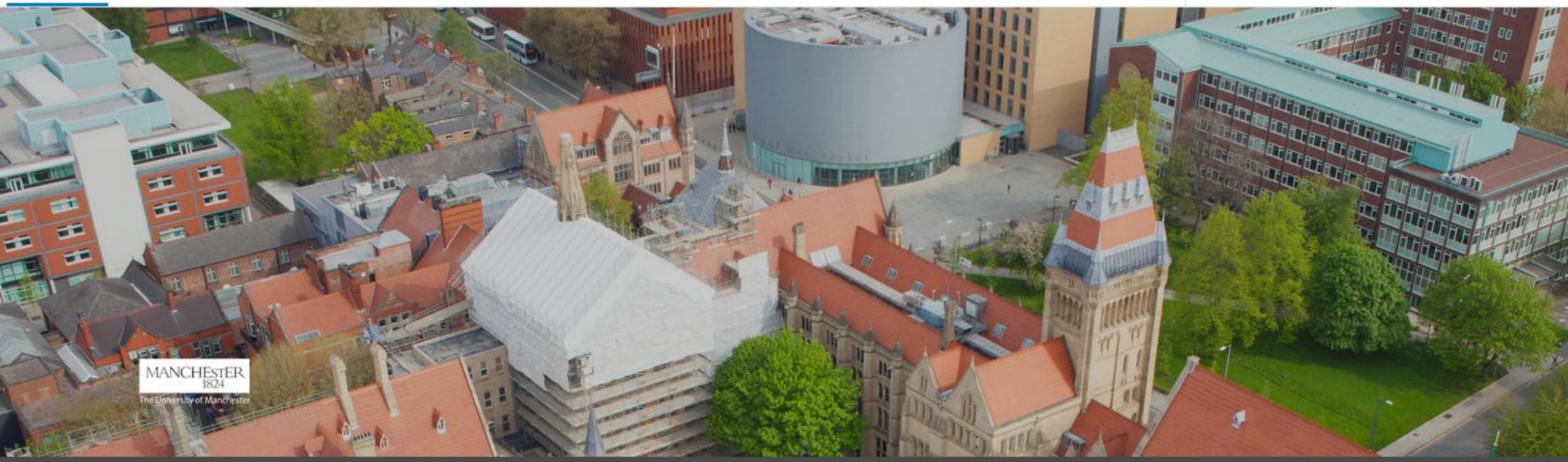
https://www.arcgis.com/index.html





Home Gallery Map Scene Groups Content Organization

Ray Burgess
ray.burgess@mancheste...

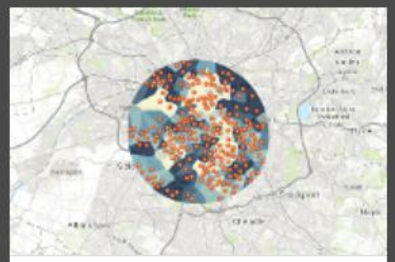


ArcGIS Online Spatial Data Infrastructure

Explore, Create and Publish Spatial Data for Teaching and Research @ The University of Manchester



Web Mapping Application
A socio-economic analysis of th...



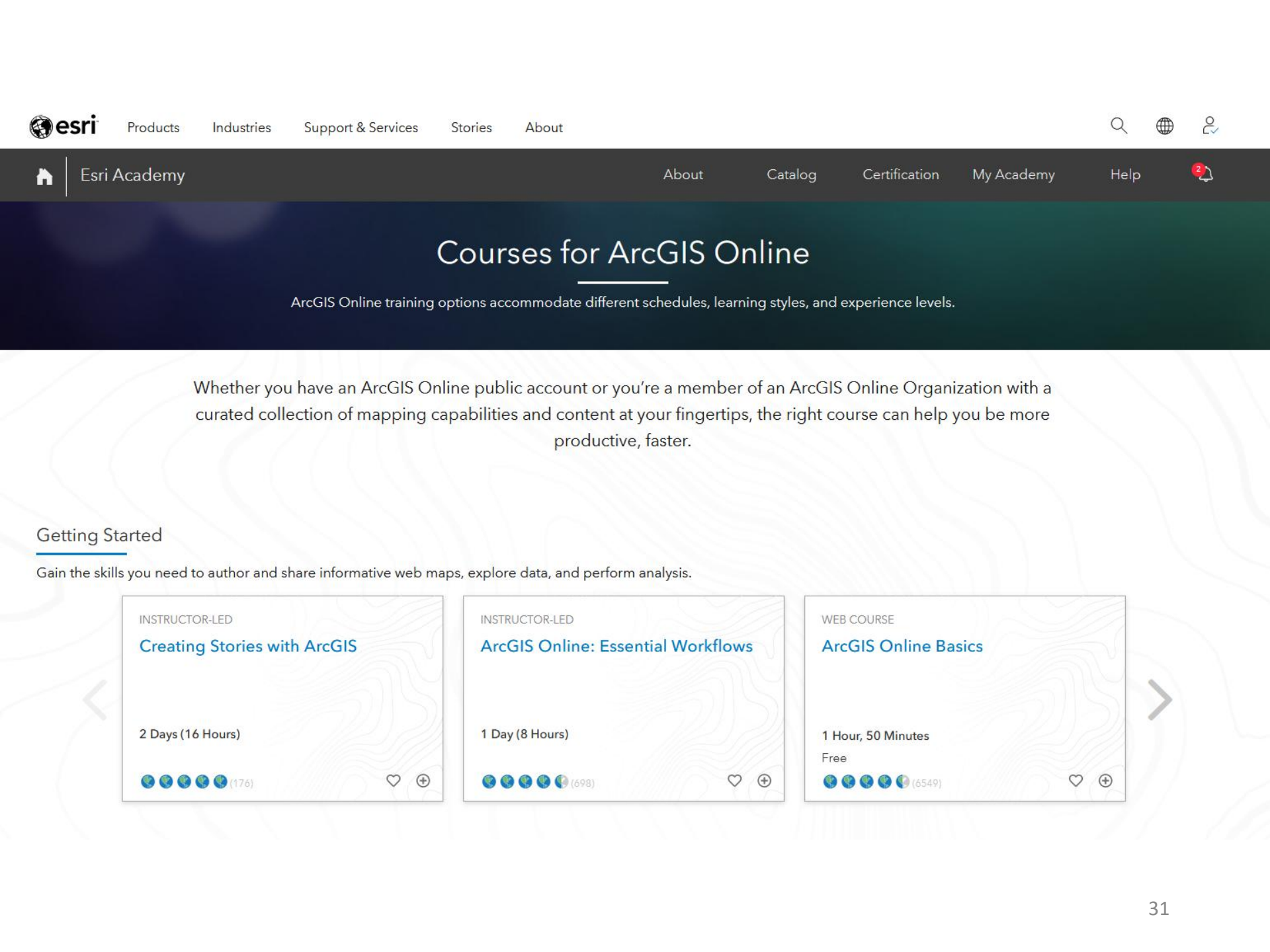
Web Mapping Application
Access to Alexandra Park, Man...



Web Map
Looper Project Population Den...



Web Map
Peak District National Park Wild...


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Stories
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Esri Academy
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Help
Courses for ArcGIS Online
ArcGIS Online training options accommodate different schedules, learning styles, and experience levels.
Whether you have an ArcGIS Online public account or you're a member of an ArcGIS Online Organization with a curated collection of mapping capabilities and content at your fingertips, the right course can help you be more productive, faster.
Getting Started
Gain the skills you need to author and share informative web maps, explore data, and perform analysis.
INSTRUCTOR-LED
Creating Stories with ArcGIS
2 Days (16 Hours)
(176)
INSTRUCTOR-LED
ArcGIS Online: Essential Workflows
1 Day (8 Hours)
(698)
WEB COURSE
ArcGIS Online Basics
1 Hour, 50 Minutes
Free
(6549)

Homework and Resources

- Read Chapter 1 of course textbook: Overview
- Take the quiz to test your basic understanding (on BB)

Further resources

ESRI training GIS Basics (components 1-3 only; 55 mins)

https://www.esri.com/training/catalog/5d9cd7de5edc347a71611ccc/gis-basics/

Have a look at Geo-Wiki and OpenStreetMap

https://www.geo-wiki.org/

https://www.openstreetmap.org/#map=6/54.910/-3.432